

ZETHETA SECURITISATION

Data Analyst Project Report

Zetheta Project 1A — Data Analyst – Securitisation

Project	Zetheta Project 1A — Data Analyst – Securitisation
Primary Tool	Microsoft Power BI
Supporting Tools	Excel, DAX Studio, CSV
Dashboard Pages	7
Validation	IFRS 9 ECL, waterfall, RLS and performance

This report documents the practical data analysis, modelling, DAX, dashboard, validation, security and performance work completed for the Zetheta securitisation portfolio project.

1. Executive Summary

The Zetheta Securitisation project is a Power BI analytical solution for monitoring an auto-loan securitisation portfolio. It combines loan, borrower, vehicle, geography and monthly performance information to support delinquency analysis, expected credit loss, stress testing, investor reporting and waterfall/trigger monitoring.

The solution follows an end-to-end Data Analyst workflow: source data preparation, modelling, DAX measure development, dashboard creation, independent Excel validation, row-level security and DAX Studio performance testing.

2. Business Problem

Business users need a consistent view of portfolio balances, delinquency, defaults, recoveries, prepayments, losses and credit risk. The dashboard converts these measures into pages designed for risk, investor and rating-oriented analysis.

3. Objectives

- Build a structured Power BI analytical model.
- Create reusable DAX measures for portfolio and risk KPIs.
- Develop business-focused dashboard pages.
- Validate key ECL outputs against Excel.
- Test representative DAX performance below the project target.
- Implement role-based row-level security.

4. Data Sources

The project uses four CSV datasets and an Excel validation workbook.

Source	Purpose
auto_loan_securitisation_data.csv	Primary loan-level portfolio data.
dpd_snapshot_history.csv	Delinquency/DPD snapshot history.
dynamic_loss_monthly.csv	Monthly dynamic loss information.
static_pool_vintage_data.csv	Static pool and vintage information.
Day9_IFRS9_ECL_Crosscheck.xlsx	Independent IFRS 9 ECL cross-check.

5. Data Model

The model uses a securitisation-oriented dimensional structure with DimLoan, DimBorrower, DimVehicle, DimGeography and DimDate dimensions, supported by FactMonthlyPerformance, FactStaticPool and FactDynamicLoss fact tables.

Table	Role
DimLoan	Loan-level descriptive attributes.
DimBorrower	Borrower attributes.
DimVehicle	Vehicle attributes.
DimGeography	Geographic attributes.
DimDate	Date analysis.
FactMonthlyPerformance	Monthly performance.
FactStaticPool	Static pool/vintage analysis.
FactDynamicLoss	Dynamic loss analysis.

6. Dashboard Structure

Page	Purpose
1. DPD Analysis	Delinquency exposure, rates, balances and loan counts.
2. Page 2	Cumulative loss/default analysis and supporting visuals.
3. IFRS 9 ECL	Expected credit loss analysis and validation.
4. Stress Testing	Recovery, prepayment, interest-rate and ECL stress impacts.
5. Investor Reporting	Portfolio KPIs and investor trends.
6. Waterfall & Trigger Report	Allocation, credit enhancement and trigger indicators.
7. Rating Agency	Vintage and cumulative performance indicators.

7. Key Analytical Measures

More than 80 validation entries were recorded across portfolio balances, collections, defaults, delinquency, ECL, prepayments, recoveries, concentration and waterfall analysis.

Area	Examples
Portfolio	Total Current Balance; Original Pool Balance; Total Loan Count
Delinquency	Loans 30+/60+/90+ DPD; DPD rates
Loss	Gross Loss YTD; Net Loss YTD; Net Loss Rate
Collections	Collection Efficiency; Recoveries YTD; Recovery Rate
Prepayments	Prepayments YTD; Monthly CPR; Monthly SMM
ECL	Total ECL Provision; ECL Provision YTD; EAD YTD
Concentration	Region Balance Concentration %; Region HHI; State HHI
Risk profile	Average CIBIL; LTV; DTI; PD; LGD

8. DPD and Delinquency Analysis

The DPD Analysis page contains KPI cards for DPD 90+ balance, DPD 90+ balance percentage, DPD 60+ rate and DPD 30+ rate, plus delinquency-bucket charts for balances and loan counts. The DPD 30+ rate was corrected using the DelinquencyDays field and validated at 14.00%.

9. IFRS 9 ECL Validation

The Power BI ECL output was cross-checked against Day9_IFRS9_ECL_Crosscheck.xlsx. The differences are at rounding/precision level.

Row	Calculated ECL	Power BI ECL	Difference
1	7,507.750575	7,507.75	0.000575
2	8,502.419539	8,502.42	-0.000461

Conclusion: the Power BI values match the Excel calculations to normal display rounding/precision.

10. Stress Testing

Scenario	Current implementation/output
Recovery stress	Recoveries reduced to 80%; Recovery Stress % = 20%.
Prepayment stress	Prepayments increased by 20%; displayed stress approximately 1.31M.
Interest-rate shock	Average InterestRate increased by 0.02; displayed rate approximately 11.03.
Stress ECL	Current Stress ECL Impact displayed approximately 4.33M.

11. Waterfall and Trigger Analysis

The Waterfall & Trigger Report presents senior/junior allocation, credit enhancement and trigger indicators. Current outputs include 70% senior allocation, 30% credit enhancement and a 0.00 allocation check. These waterfall assumptions are analytical/illustrative and should not be interpreted as contractual terms.

Indicator	Output
Senior Tranche Allocation %	70%
Junior Tranche Allocation %	30%
Credit Enhancement %	30%
Waterfall Allocation Check	0.00
Default Rate Trigger	0
Excess Spread Trigger	0

12. Row-Level Security

Five roles were configured and validated: Servicer Viewer (SVC001), Servicer Viewer 2 (SVC002), Risk Analyst (SVC003), Investor Viewer (SVC004) and Auditor (SVC005). The implementation is a static demonstration and is not a dynamic USERPRINCIPALNAME design.

13. DAX Studio Performance

DAX Studio Server Timings was used for representative KPI testing. The final representative query returned the selected KPI set in 13 ms, below the project target of less than 100 ms.

Test	Result
Representative KPI query	13 ms
Target	<100 ms
Status	PASS

The tested KPI set included Total Current Balance, Total Loan Count, Total ECL Provision, Total Net Loss and Total Recoveries.

14. Investor and Rating Reporting

Investor Reporting contains current balance, original pool balance, overdue amount, delinquent loans, SMM, recovery rate, defaults, gross loss, net loss, recoveries, prepayments and CPR. The Rating Agency page contains cumulative loss, default, recovery and prepayment measures with vintage-level information.

15. Visual Design and Usability

The report uses the Zetheta Securitisation theme with navy/blue colours, risk-oriented colours and consistent typography. Card labels across all seven pages were formatted consistently with bold text and navy colour. A Page Navigator provides navigation between the seven pages.

16. Key Results Snapshot

Metric	Validated/current output
Total Current Balance	317,785,202.14
Original Pool Balance	546,073,700
Total Loan Count	500
Default Loan Count	9
Default Rate	1.8%
Defaulted Balance	6,789,682.49
Total ECL Provision	11,662,952.90
Total Net Loss	879,039
Total Recoveries	396,589
DAX Studio test	13 ms

17. Tools and Technologies

- Microsoft Power BI Desktop — modelling and dashboard development.
- DAX — business logic and measures.
- DAX Studio — performance testing.
- Microsoft Excel — ECL validation.
- CSV — source data ingestion.

18. Limitations and Future Improvements

- Replace illustrative waterfall assumptions with verified transaction-specific terms when available.
- Use dynamic USERPRINCIPALNAME-based RLS for production deployment where appropriate.
- Expand automated validation between Power BI and independent calculations.
- Complete formal measure documentation and final submission packaging.

19. Conclusion

The Zetheta Securitisation project demonstrates an end-to-end Data Analyst workflow covering data preparation, dimensional modelling, DAX, dashboard design, risk analysis, IFRS 9 ECL validation, stress testing, RLS and performance testing. The completed dashboard provides a structured portfolio-monitoring solution suitable for demonstrating analytical and business-intelligence skills.

Appendix — File Inventory

Location	Item
PowerBi	Zetheta_Project_1A.pbix
Excel	Day9_IFRS9_ECL_Crosscheck.xlsx
Data	auto_loan_securitisation_data.csv
Data	dpd_snapshot_history.csv
Data	dynamic_loss_monthly.csv
Data	static_pool_vintage_data.csv
Project PDF/report	This report
Presentation	To be populated
GitHub	To be populated if required

End of Report